Interrogation of brain in patients with severe COVID-19

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Introduction

Coronavirus disease 2019 (COVID-19) outbreak has quickly evolved into an urgent health care crisis worldwide. Due to typical clinical manifestations affect the respiratory system in affected patients, many early studies overlook COVID-19s affect on neurological and cognitive function. The evidence for short- and long- lasting neurological impact of COVID-19 is increasing with wide ranges of effects on symptoms, duration, and recovery of cognitive functions. Central Nervous system dysfunctions (delirium, psychomotor retardation, ataxia, stroke) that are observed during and after acute phase become more commonly recognized. [1] Thus, it has become pertinent to study underlying mechanisms of severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) on multiple loci in the brain.

Here we aim to understand the causes of neurological dysfunction and which molecular processes are affected in brains of COVID-19 patients in a systematic and unbiased way using single-nucleus RNA sequencing of 3 different types of brain tissue: prefrontal cortex, medulla oblongata and choroid plexus.[2] Prefrontal cortex (PFC) is involved in higher order cognitive function. Medulla oblongata (medulla) is the control center for respiratory system. Choroid plexus (CP) provides is involved in immune response in the brain. Comparing the different transcriptomic landscape of severe COVID-19 patients and the control group would shed light into how brain response to SARS-CoV-2 and identify important markers.

Results

Transcriptome profiling of brain regions of COVID-19 patients and controls

Careful characterization of transcriptomic profile of CNS of COVID-19 patients is carried out by droplet base single-nucleus RNA sequencing in PFC, medulla, and CP in 5 patients and 4 controls. The data was preprocessed to obtain 68557 high-quality single-nucleus barcodes with total of 20636 detected genes. Nuclei with fewer than 200 genes or more than 5% mitochondrial reads were removed during preprocessing step.

Shared nearest neighbor graph clustering was performed on dimensionally reduced data using principal component analysis where 13 clusters were identified (Fig1a). Additionally, uniform manifold approximation and projection (UMAP) was performed across all brain

regions and samples. COVID-19 status, subject or tissue does not agree with annotated clusters, but the identified clusters correspond well with cell type (Fig1b-e).

Differential gene expression analysis shows brain immune response of COVID-19 patients

Differential gene expression analysis between clusters was carried out to identify marker genes. Subsequently functional analysis using DAVID was performed.[3] Functional analysis show high concordance of human brain tissue, peripheral blood cells, and brain organoids suggesting robust clusters of cell population among the samples.

We next examine genes that are differentially expressed between COVID-19 patients. 50 genes that were differentially expressed were identified and functional enrichment analysis was carried out using DAVID. The annotation clusters identified that the top 20 differentially expressed gene is involved in viral transcription(17 genes, p-value = 9.4E-24), calcium binding(13 genes, p-value = 9.7E-7), and many immune responses including innate immune response(6 genes, p-value = 7.5E-3), T cell receptor signaling(5 genes, p-value= 8.7E-3), and antigen processing(5 genes, p-value = 4.3E-4). The most differentially expressed gene on average between the two group was TYROBP gene with average log2FC of 0.6846 (adjusted p-value = 7.605525e-213). TYROBP gene encodes for tyrosine kinase binding protein and is found in variety of cells produced in the bone marrow and other immune cells. The TYROBP interacts with proteins and plays an important role in microglia, glial cells that function as macrophages in CNS, and has been previously described to have increased inflammatory response in COVID-19 cases.[4]

Comparing Fig. 1f and 1b, TYROBP gene dysregulation seems occurs in microglial cells (Mic) and macrophages (Mo) clusters. Further exploring the expression levels of TYROBP, expression levels of the brain regions and tissue types are shown in Fig2. Medulla is shown with drastic dysregulation which may not be surprising as SARS-CoV-2 mainly attacks the respiratory system. Interestingly, when comparing COVID patients with control in Mic and Mo cells, we observe expression level change mainly focused on the microglial cells (Mic). Overall, this recapitulates that immune response have deteriorating effect on COVID-19 patients in the brain. Previous study has shown that multiple stresses to the microglia may lead to more severe onset of COVID-19. [6]

Discussion

As the world start its step into a “with COVID” stage of the pandemic, it is important to understand effects the virus has on patient’s neurological function. Some effects may remain long term while some subside. Understanding of the mechanism and determining factors of severity and disease state is important if we wish to possibly reverse the effect that SARS-CoV-2 has had on patients. The advancement in sequencing technology has

provided unprecedented details to the transcriptomic landscapes. To understand the effects of the COVID-19 on patients' neurological function, we investigated individual cells across 3 different brain regions: prefrontal cortex, choroid plexus, and medulla oblongata. Here after careful study, we find that genes related to immune response is dysregulated in COVID-19 patients. TYROBP, the highest dysregulated gene, is upregulated in patients and is involved in microglial immune response. As dysregulation mainly occurs in microglial cells, this gene can provide marker for microglial cells in patients. As cells were collected from deceased patients, we may infer TYROBP maybe indicator of severity of the virus. Also, this result may suggest potential targeted treatment for microglial may be beneficial to mitigate the severity of COVID-19.

Materials and methods

Data

The brain tissue samples were obtained from 5 COVID-19 patients and 4 controls. From each donor, samples from 3 different brain regions (PFC, medulla, CP) were collected. Single-nucleus RNA-seq (snRNA-seq) was performed. This sequencing technique is employed for complex tissue RNA profiling. In snRNA-seq, nuclei is isolated and introduced to microfluidic device that encapsulates single nuclei with barcoded mRNA-capture beads in a droplet. Here RNA is reverse transcribed to cDNA, amplified, and finally sequenced.[7] Raw data is available on NCBI GEO(GSE164485).

Analysis

Principle component analysis is performed on the preprocessed data and 100 principal components were identified. Using the first 20 components, shared nearest neighbor clustering was performed. To visualize the results, uniform manifold approximation and projection visualization was employed. Differential expression is calculated based on non-parametric Wilcoxon rank sum test between specified groups. We used Seurat to perform normalization, PCA and UMAP dimensionality reduction and visualization, and to calculated differential gene expression.[5] DAVID was used to carry out gene set enrichment analysis on identified differentially expressed genes.[3]

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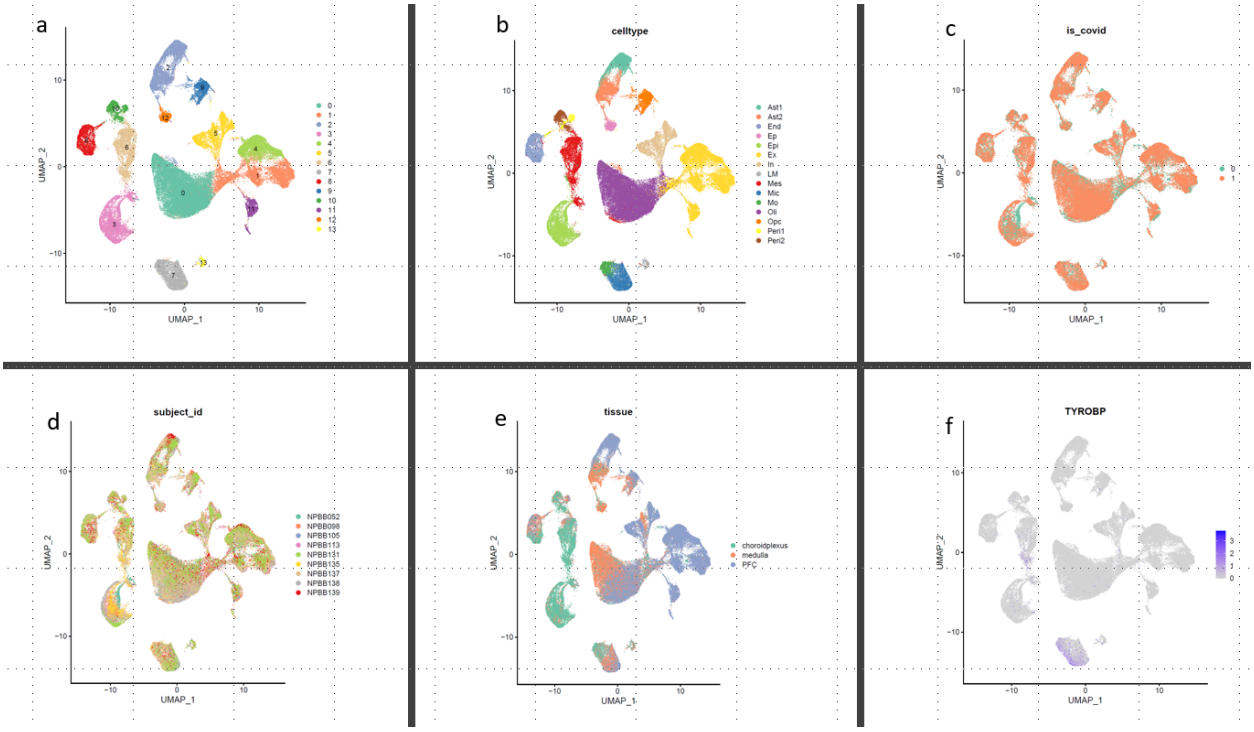


Fig. 1 Uniform manifold approximation and projection visualization of 10X RNA sequencing data. Shared nearest neighbor clustering identifies 13 clusters(a) which corresponds to the cell type(b). Other factors such as COVID status(c), subject id(d), and tissue (e) are also shown. TYROBP is shown in the population of Mic and Mo cluster(f).

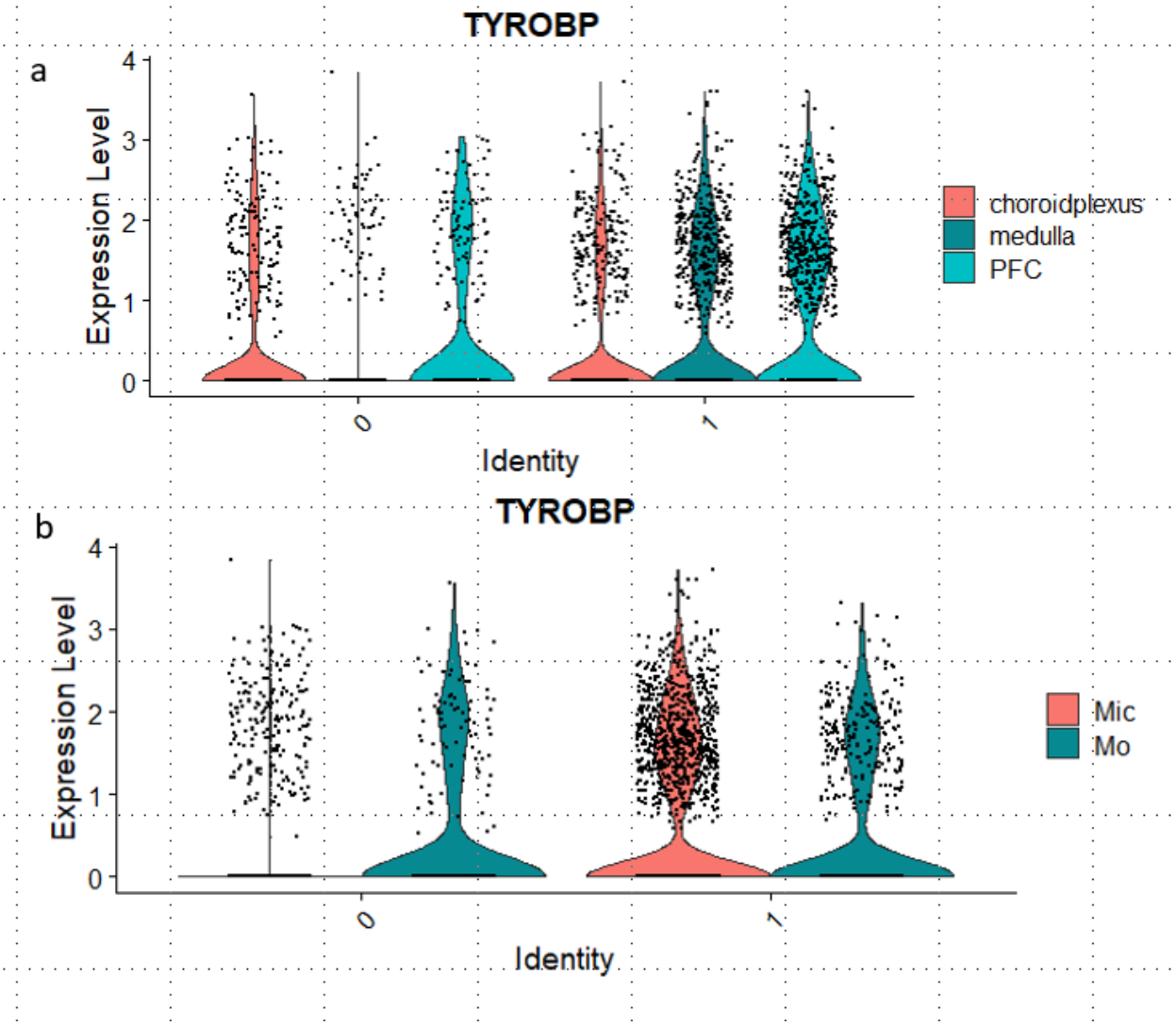


Fig. 2 Dysregulation of TYROBP in COVID-19 patients. Expression levels of TYROBP is shown by tissue type(a) and cell type(b).